Installing App with Helm

MANAGING KUBERNETES WITH HELM

Installing first app with Helm

Understanding Apache

The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

Deploying this apache chart will create these resources:

- **Services** for networking
- **Deployment** for deploying apache
- **ReplicaSet** for creating pods
- **Pods** for running apache process

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Understanding Apache

Creating these Kubernetes resources requires both Apache and Kubernetes expertise.

Apache expertise is required because the user needs to know the required physical components, as well as how to configure them.

Kubernetes expertise is required because users need to know how to deploy Apache dependencies as Kubernetes resources. Given the complexity and number of components that are required, deploying Apache on Kubernetes is a task.

Rather than focusing on creating and configuring each of the Kubernetes resources we have described, users can leverage Helm as a package manager to deploy and configure Apache on Kubernetes. To begin, explore a platform called **Artifact Hub** to locate a suitable Apache Helm chart.

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Finding Apache Chart

Helm charts can be made available for consumption by publishing them to a chart repository. A chart repository is a location where packaged charts can be stored and shared. Artifact Hub is a centralized location for upstream Kubernetes artifacts, such as Helm charts, operators, plugins, and more.

To search for charts in Artifact Hub, use:
helm search hub

Once a repository has been added, you can search across these repositories. using:
helm search repo

To locate charts containing the apache keyword:
helm search hub apache

```
alok@workstation:~$
alok@workstation:~$ helm search hub apache
URL                                                    CHART VERSION        APP VERSION        DESCRIPTION

https://artifacthub.io/packages/helm/bitnami/ap...     11.0.3               2.4.59             Apache HTTP S
er is an open-source HTTP serve...
https://artifacthub.io/packages/helm/bitnami-ak...     9.2.7                2.4.54             Apache HTTP S
er is an open-source HTTP serve...
https://artifacthub.io/packages/helm/camptocamp...     0.4.0                1.0                A Helm chart
 Kubernetes
```

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Finding Apache Chart

Run the following command by including the --max-col-width flag to view the untruncated results in tabular format:
helm search hub apache --max-col-width=0

Pass the --output flag and specify either yaml or json, which will print the search results in full.
helm search hub apache --output yaml

```
alok@workstation:~$
alok@workstation:~$
alok@workstation:~$ helm search hub apache --output yaml
- app_version: 2.4.59
  description: Apache HTTP Server is an open-source HTTP server. The goal of this
    project is to provide a secure, efficient and extensible server that provides
    HTTP services in sync with the current HTTP standards.
  repository:
    name: bitnami
    url: https://charts.bitnami.com/bitnami
  url: https://artifacthub.io/packages/helm/bitnami/apache
  version: 11.0.3
- app_version: 2.4.54
  description: Apache HTTP Server is an open-source HTTP server. The goal of this
    project is to provide a secure, efficient and extensible server that provides
    HTTP services in sync with the current HTTP standards.
```

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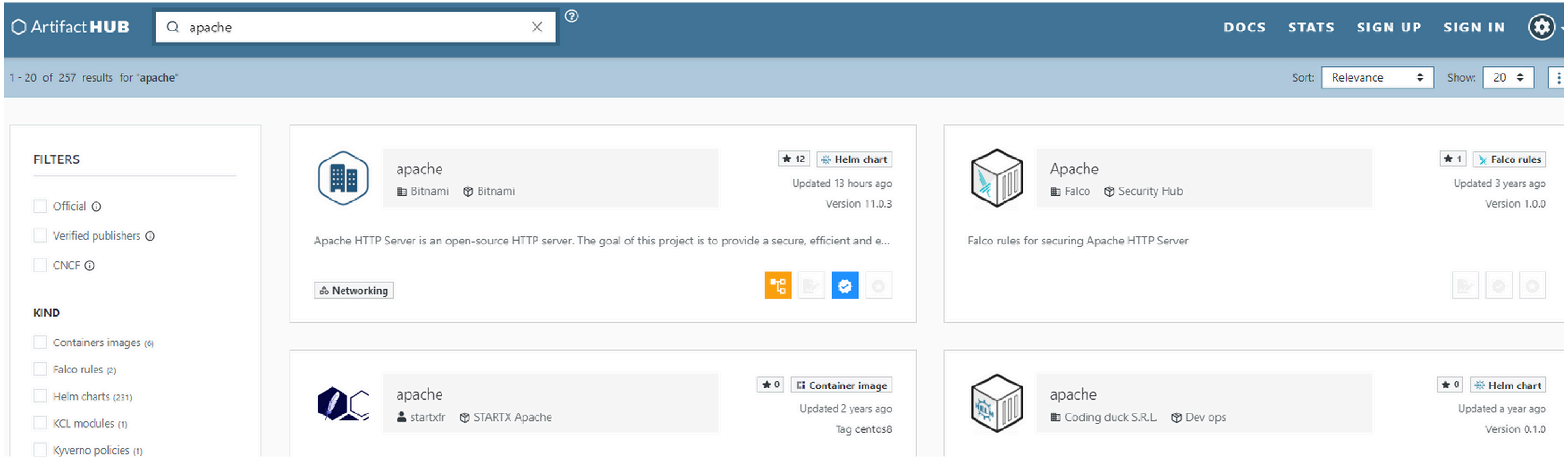
Finding Apache Chart using Browser



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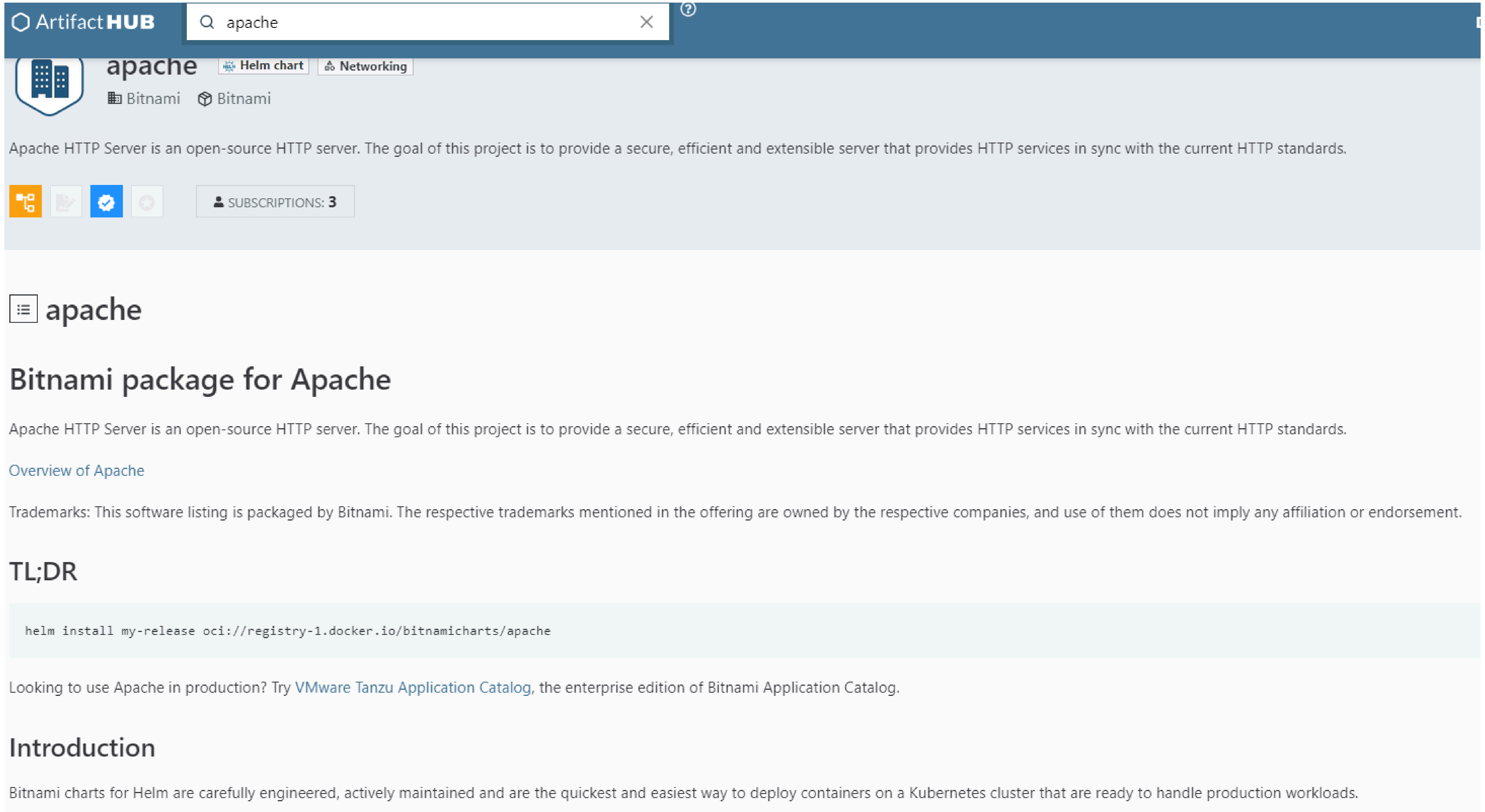
Finding Apache Chart using Browser

A chart's Artifact Hub page can provide this URL, along with other installation details. Once you have entered the Apache chart's URL into a browser window, a page similar to the following will be displayed:



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Adding Bitnami repository

A chart's Artifact Hub page can provide this URL, along with other installation details. Under the TL;DR heading, you should see a helm repo add command. This is the command that you need to run to add the Bitnami chart repository, which is the repository that contains the Apache chart we are interested in installing.

Add the full Bitnami chart repository
helm repo add bitnami https://charts.bitnami.com/bitnami

Verify that the chart has been added:
helm repo list

```
alok@workstation:~$
alok@workstation:~$ helm repo list
NAME                    URL
sumologic               https://sumologic.github.io/sumologic-kubernetes-collection
prometheus-community    https://prometheus-community.github.io/helm-charts
alok@workstation:~$
alok@workstation:~$ helm repo add bitnami https://charts.bitnami.com/bitnami
"bitnami" has been added to your repositories
alok@workstation:~$
alok@workstation:~$ helm repo list
NAME                    URL
sumologic               https://sumologic.github.io/sumologic-kubernetes-collection
prometheus-community    https://prometheus-community.github.io/helm-charts
bitnami                 https://charts.bitnami.com/bitnami
alok@workstation:~$
alok@workstation:~$
```

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Adding Bitnami repository

View charts from locally configured repositories that contain the bitnami keyword
helm search repo bitnami --output yaml

```
alok@workstation:~$
alok@workstation:~$ helm search repo bitnami --output yaml
- app_version: 2.9.1
  description: Apache Airflow is a tool to express and execute workflows as directed
    acyclic graphs (DAGs). It includes utilities to schedule tasks, monitor task progress
    and handle task dependencies.
  name: bitnami/airflow
  version: 18.0.8
- app_version: 2.4.59
  description: Apache HTTP Server is an open-source HTTP server. The goal of this
    project is to provide a secure, efficient and extensible server that provides
    HTTP services in sync with the current HTTP standards.
  name: bitnami/apache
  version: 11.0.2
```

Check you have access to the Apache chart, run the following helm search repo command with the apache argument:
helm search repo apache

```
alok@workstation:~$
alok@workstation:~$ helm search repo apache
NAME                          CHART VERSION   APP VERSION   DESCRIPTION

bitnami/apache                11.0.3          2.4.59        Apache HTTP Server is an open-sourc
P serve...
bitnami/airflow               18.0.10         2.9.1         Apache Airflow is a tool to express
execute...
bitnami/apisix                3.0.5           3.9.1         Apache APISIX is high-performance,
```

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Installing Apache Chart

Previous versions can be queried by passing the --versions flag to the search command:
helm search repo apache --versions

```
alok@workstation:~$
alok@workstation:~$ helm search repo apache --versions
NAME                                          CHART VERSION   APP VERSION    DESCRIPTION

bitnami/apache                                11.0.3          2.4.59         Apache HTTP Server is an o
P serve...
bitnami/apache                                11.0.2          2.4.59         Apache HTTP Server is an o
P serve...
bitnami/apache                                11.0.1          2.4.59         Apache HTTP Server is an o
P serve
```

You can find a lot of important details about a Helm chart with the following helm show subcommands:

helm show chart bitnami/apache --version 11.0.3
helm show readme bitnami/apache --version 11.0.3
helm show values bitnami/apache --version 11.0.3
helm show all bitnami/apache --version 11.0.3

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Installing Apache Chart

If we explore the helm show values output further, we can find values that pertain to configuring the Apache's metadata:

```
## @param affinity Affinity for pod assignment
## Ref: https://kubernetes.io/docs/concepts/configuration/assign-pod-node/#affinity-anc
## NOTE: podAffinityPreset, podAntiAffinityPreset, and nodeAffinityPreset will be ignor
##
affinity: {}
## @param nodeSelector Node labels for pod assignment
## ref: https://kubernetes.io/docs/concepts/scheduling-eviction/assign-pod-node/
##
nodeSelector: {}
## @param tolerations Tolerations for pod assignment
## ref: https://kubernetes.io/docs/concepts/configuration/taint-and-toleration/
##
tolerations: []
## @param topologySpreadConstraints Topology Spread Constraints for pod assignment spre
ure-domains. Evaluated as a template
## Ref: https://kubernetes.io/docs/concepts/workloads/pods/pod-topology-spread-constra
##
topologySpreadConstraints: []
## @param extraPodSpec Optionally specify extra PodSpec
##
extraPodSpec: {}
## Get the server static content from a git repository
## @param cloneHtdocsFromGit.enabled Get the server static content from a git reposito
## @param cloneHtdocsFromGit.repository Repository to clone static content from
## @param cloneHtdocsFromGit.branch Branch inside the git repository
## @param cloneHtdocsFromGit.enableAutoRefresh Enables an automatic git pull with a sic
## @param cloneHtdocsFromGit.interval Interval for sidecar container pull from the repo
```

Installing first app with Helm

Installing Apache Chart, with custom values

Let's override them by creating a values file. Create a new file on your machine called **apache-custom.yaml**. In that file, enter the following content:

```
commonLabels:
  managed-by: helm
nodeSelector:
  mem: large
resources:
  limits:
    cpu: 200m
    memory: 200Mi
  requests:
    cpu: 100m
    memory: 100Mi
service:
  type: NodePort
```

Installing first app with Helm

Installing Apache Chart

Create a namespace for our application:
kubectl create ns apache

We use helm install to install a Helm chart. The standard syntax is as follows:
helm install [NAME] [CHART] [flags]

helm install apache bitnami/apache --values=apache-custom.yaml --namespace apache --version 11.0.3

Installing first app with Helm

Installing Apache Chart

If the chart's installation is successful, you should see the following output:

```
NAME: apache
LAST DEPLOYED: Tue May 14 15:02:39 2024
NAMESPACE: apache
STATUS: deployed
REVISION: 1
TEST SUITE: None
NOTES:
CHART NAME: apache
CHART VERSION: 11.0.3
APP VERSION: 2.4.59

** Please be patient while the chart is being deployed **

1. Get the Apache URL by running:

  export NODE_PORT=$(kubectl get --namespace apache -o jsonpath="{.spec.ports[0].nodePort}" services apache)
  export NODE_IP=$(kubectl get nodes --namespace apache -o jsonpath="{.items[0].status.addresses[0].address}")
  echo http://$NODE_IP:$NODE_PORT/


WARNING: You did not provide a custom web application. Apache will be deployed with a default page. Check the READ
tion "Deploying your custom web application" in https://github.com/bitnami/charts/blob/main/bitnami/apache/README.
loying-a-custom-web-application.
alok@workstation:~$ _
```

Installing first app with Helm

Inspecting Application Release

One of the easiest ways to inspect a release and verify its installation is to list all the Helm releases in a given namespace

helm list --namespace apache

```
alok@workstation:~$
alok@workstation:~$ helm list --namespace apache
NAME     NAMESPACE        REVISION         UPDATED                                  STATUS          CHART
ION
apache   apache           1                2024-05-14 15:02:39.650121955 +0530 IST deployed        apache-11.0.3
```

Helm provides the **get** subcommand to provide more information about a release.

helm get hooks apache --namespace apache
helm get values apache --namespace apache
helm get manifest apache --namespace apache
helm get notes apache --namespace apache
helm get all apache --namespace apache

Installing first app with Helm

Accessing Apache

Run the command and paste the output on terminal to get node IP and port

helm get notes apache --namespace apache

```
alok@workstation:~$
alok@workstation:~$ helm get notes apache --namespace apache
NOTES:
CHART NAME: apache
CHART VERSION: 11.0.3
APP VERSION: 2.4.59

** Please be patient while the chart is being deployed **

1. Get the Apache URL by running:

  export NODE_PORT=$(kubectl get --namespace apache -o jsonpath="{.spec.ports[0].nodePort}" services apache)
  export NODE_IP=$(kubectl get nodes --namespace apache -o jsonpath="{.items[0].status.addresses[0].address}")
  echo http://$NODE_IP:$NODE_PORT/


WARNING: You did not provide a custom web application. Apache will be deployed with a default page. Check the REA
tion "Deploying your custom web application" in https://github.com/bitnami/charts/blob/main/bitnami/apache/READMI
loying-a-custom-web-application.
```

Installing first app with Helm

Accessing Apache

Check the label of pods, services. You will find the label - managed-by: helm.
Also check “nodeSelector” in pod and check your “custom-apache.yaml” file to relate.
Do check the resource block in the pod

```
alok@workstation:~$
alok@workstation:~$ kubectl -n apache get pods --show-labels
NAME                      READY   STATUS    RESTARTS   AGE   LABELS
apache-5d56d674bb-7x2jd   1/1     Running   0          13m   app.kubernetes.io/instance=apache,app.kubernetes.io/managed
-by=Helm,app.kubernetes.io/name=apache,app.kubernetes.io/version=2.4.59,helm.sh/chart=apache-11.0.3,managed-by=helm,pod-
template-hash=5d56d674bb
alok@workstation:~$
alok@workstation:~$ kubectl -n apache get svc --show-labels
NAME     TYPE       CLUSTER-IP      EXTERNAL-IP   PORT(S)                      AGE   LABELS
apache   NodePort   10.104.70.159   <none>        80:31661/TCP,443:32665/TCP   13m   app.kubernetes.io/instance=apache,a
pp.kubernetes.io/managed-by=Helm,app.kubernetes.io/name=apache,app.kubernetes.io/version=2.4.59,helm.sh/chart=apache-11.
0.3,managed-by=helm
alok@workstation:~$
```

Installing first app with Helm

Accessing Apache

Get the NodePort of the service using:

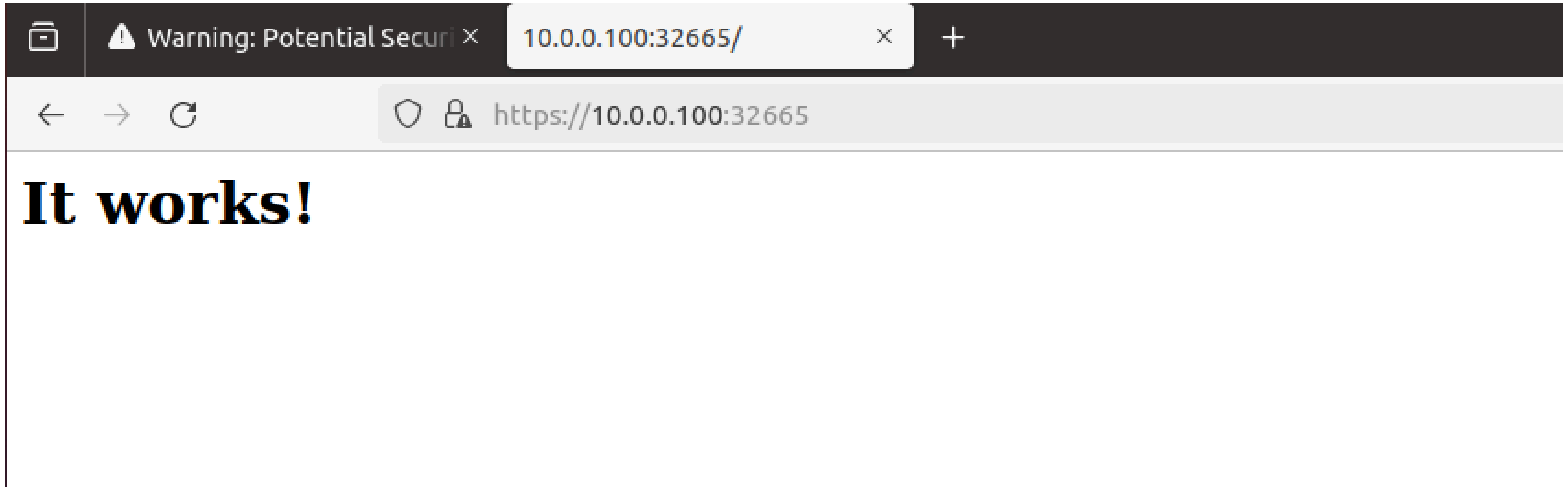
kubectl -n apache describe svc apache

```
alok@workstation:~$ kubectl  -n apache describe svc apache
Name:                     apache
Namespace:                apache
Labels:                   app.kubernetes.io/instance=apache
                          app.kubernetes.io/managed-by=Helm
                          app.kubernetes.io/name=apache
                          app.kubernetes.io/version=2.4.59
                          helm.sh/chart=apache-11.0.3
                          managed-by=helm
Annotations:              meta.helm.sh/release-name: apache
                          meta.helm.sh/release-namespace: apache
Selector:                 app.kubernetes.io/instance=apache,app.kubernetes.io/name=apache
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.104.70.159
IPs:                      10.104.70.159
Port:                     http  80/TCP
TargetPort:               http/TCP
NodePort:                 http  31661/TCP
Endpoints:                172.16.3.189:8080
Port:                     https  443/TCP
TargetPort:               https/TCP
NodePort:                 https  32665/TCP
Endpoints:                172.16.3.189:8443
Session Affinity:         None
External Traffic Policy:  Cluster
Events:                   <none>
```

Installing first app with Helm

Accessing Apache

Now go to workstation machine and access your apache using https://node-ip:node-port



Upgrading app with Helm

Upgrading Apache Release

Upgrading a release refers to the process of modifying the release's values or updating the chart to a newer version. Add these modifications to the apache-custom.yaml file.

```
commonLabels:
  managed-by: helm
nodeSelector:
  mem: large
resources:
  limits:
    cpu: 200m
    memory: 200Mi
  requests:
    cpu: 100m
    memory: 100Mi
service:
  type: NodePort
replicaCount: 2
```

Upgrading app with Helm

Running Upgrade

Upgrading a release refers to the process of modifying the release's values or updating the chart to a newer version. Add these modifications to the apache-custom.yaml file.

```
commonLabels:
  managed-by: helm
nodeSelector:
  mem: large
resources:
  limits:
    cpu: 200m
    memory: 200Mi
  requests:
    cpu: 100m
    memory: 100Mi
service:
  type: NodePort
replicaCount: 2
```

Upgrading app with Helm

Running Upgrade

Run the helm upgrade command. You should see an output similar to that of helm install. You should also notice that the REVISION field now says 2.

helm upgrade apache bitnami/apache --values apache-custom.yaml -n apache --version 11.0.3

```
alok@workstation:~$
alok@workstation:~$ helm upgrade apache bitnami/apache --values apache-custom.yaml -n apache --version 11.0.3
Release "apache" has been upgraded. Happy Helming!
NAME: apache
LAST DEPLOYED: Tue May 14 15:26:38 2024
NAMESPACE: apache
STATUS: deployed
REVISION: 2
TEST SUITE: None
NOTES:
CHART NAME: apache
CHART VERSION: 11.0.3
APP VERSION: 2.4.59

** Please be patient while the chart is being deployed **

1. Get the Apache URL by running:

  export NODE_PORT=$(kubectl get --namespace apache -o jsonpath="{.spec.ports[0].nodePort}" services apache)
  export NODE_IP=$(kubectl get nodes --namespace apache -o jsonpath="{.items[0].status.addresses[0].address}")
  echo http://$NODE_IP:$NODE_PORT/


WARNING: You did not provide a custom web application. Apache will be deployed with a default page. Check the READ
tion "Deploying your custom web application" in https://github.com/bitnami/charts/blob/main/bitnami/apache/README.
loying-a-custom-web-application.
```

Upgrading app with Helm

Running Upgrade

You should now see two pods running.

```
alok@workstation:~$
alok@workstation:~$ kubectl -n apache get pods
NAME                        READY   STATUS    RESTARTS   AGE
apache-5d56d674bb-46hzf     1/1     Running   0          2m33s
apache-5d56d674bb-7x2jd     1/1     Running   0          26m
alok@workstation:~$
alok@workstation:~$
```

In Kubernetes, new pods are created when their pod template is modified. The same behavior can be observed in Helm. The values that were added during the upgrade introduced a configuration change to the Apache pod template. As a result, new Apache pods were created with the updated configuration. These changes can be observed using the same “helm get manifest apache -n apache” and “kubectl get deployment -n apache” commands.

Resetting & Reusing Values

Reusing and resetting values during an upgrade

Let's look at these flags now:

- --reuse-values: When upgrading, reuse the last release's values
- --reset-values: When upgrading, reset the values to the chart defaults

If an upgrade is performed without providing values with the --set or --values flags, then the --reuse-values flag is applied by default. In other words, the same values that were used by the previous release will be used again during the upgrade if no values are provided. Alternatively, if at least one value is provided with --set or --values, then the --reset-values flag is applied by default.

Run another upgrade command without specifying any values and observe.

```
# helm upgrade apache bitnami/apache -n apache --version 11.0.3
```

Rollback

Rollback release

The helm rollback command is used to go back to previous version / revision.

Every Helm release has a history of revisions. A revision is used to track the values, Kubernetes resources, and the chart version that were used in a particular release version. A new revision is created when a chart is installed, upgraded, or rolled back. Revision data is saved in Kubernetes Secrets by default (other options are ConfigMaps, local memory, or a PostgreSQL database, as determined by the HELM_DRIVER environment variable). This allows your Helm release to be managed and interacted with by different users on the Kubernetes cluster.

Each of these Secrets corresponds with an entry of the release's revision history, which can be viewed by running the helm history command:

```
alok@workstation:~$ kubectl -n apache get secrets
NAME                           TYPE                 DATA   AGE
sh.helm.release.v1.apache.v1   helm.sh/release.v1   1      39m
sh.helm.release.v1.apache.v2   helm.sh/release.v1   1      15m
alok@workstation:~$
alok@workstation:~$ helm history apache -n apache
REVISION        UPDATED                         STATUS          CHART           APP VERSION     DESCRIPTION
1               Tue May 14 15:02:39 2024        superseded      apache-11.0.3   2.4.59          Install complete
2               Tue May 14 15:26:38 2024        deployed        apache-11.0.3   2.4.59          Upgrade complete
alok@workstation:~$
alok@workstation:~$
```

Rollback

Rollback release

Revisions that have a status of **superseded** are no longer up to date, while the revision that says **deployed** is the currently deployed revision. Other statuses include **pending** and pending_upgrade, which means the installation or upgrade is currently in progress. **failed** refers to a particular revision that has failed to install or be upgraded and unknown means that you encountered a bug and may want to file an issue or notify the maintainers. It's unlikely you will ever encounter a release with a state of unknown.

The helm get commands described previously can be used against a revision number by specifying the --revision flag.

Rollback

Rollback release

```
alok@workstation:~$ helm -n apache get values apache --revision 1
USER-SUPPLIED VALUES:
commonLabels:
  managed-by: helm
nodeSelector:
  mem: large
resources:
  limits:
    cpu: 200m
    memory: 200Mi
  requests:
    cpu: 100m
    memory: 100Mi
service:
  type: NodePort
alok@workstation:~$ helm -n apache get values apache --revision 2
USER-SUPPLIED VALUES:
commonLabels:
  managed-by: helm
nodeSelector:
  mem: large
replicaCount: 2
resources:
  limits:
    cpu: 200m
    memory: 200Mi
  requests:
    cpu: 100m
    memory: 100Mi
service:
  type: NodePort
alok@workstation:~$
```

Rollback

Running Rollback

The helm rollback command has the following syntax: helm rollback <RELEASE> [REVISION] [flags]

helm rollback apache 1 -n apache

```
alok@workstation:~$ helm history apache -n apache
REVISION        UPDATED                         STATUS          CHART           APP VERSION     DESCRIPTION
1               Tue May 14 15:02:39 2024        superseded      apache-11.0.3   2.4.59          Install complete
2               Tue May 14 15:26:38 2024        deployed        apache-11.0.3   2.4.59          Upgrade complete
alok@workstation:~$
alok@workstation:~$ helm rollback apache 1 -n apache
Rollback was a success! Happy Helming!
alok@workstation:~$
alok@workstation:~$
alok@workstation:~$ kubectl -n apache get pods
NAME                      READY   STATUS    RESTARTS   AGE
apache-5d56d674bb-7x2jd   1/1     Running   0          47m
alok@workstation:~$
alok@workstation:~$ helm history apache -n apache
REVISION        UPDATED                         STATUS          CHART           APP VERSION     DESCRIPTION
1               Tue May 14 15:02:39 2024        superseded      apache-11.0.3   2.4.59          Install complete
2               Tue May 14 15:26:38 2024        superseded      apache-11.0.3   2.4.59          Upgrade complete
3               Tue May 14 15:50:19 2024        deployed        apache-11.0.3   2.4.59          Rollback to 1
alok@workstation:~$
alok@workstation:~$
```

Uninstalling

Uninstalling Apache

helm uninstall apache -n apache
helm list -n apache
kubectl delete ns apache

```
alok@workstation:~$
alok@workstation:~$ helm uninstall apache -n apache
release "apache" uninstalled
alok@workstation:~$
alok@workstation:~$ helm list -n apache
NAME    NAMESPACE       REVISION        UPDATED STATUS  CHART   APP VERSION
alok@workstation:~$
alok@workstation:~$ kubectl delete ns apache
namespace "apache" deleted
alok@workstation:~$
alok@workstation:~$
```